

Face-2-Face with practitioner

“There are constant innovations that are required and hence there is constant scope for growth. For example, when the iPhone was launched, other companies wanted to make phones that could provide even more functions than the iPhone and yet have an equally eye-catching interface”

Sumit Sharma
Design engineer
Samsung India

How would you describe your profession? Would you say that it is fairly representative of a career in embedded design?

Yes, it is definitely representative of product-based embedded design. I work with Samsung India and we basically work to improve the software available on the phone. Every six months we get a new project, a new aspect to be worked on, and that keeps me going. Similarly, there are other phone companies, core hardware, system design, electronics, medical equipment manufacturers that also work towards improving their software and have their product engineering departments. But every company has its own requirements and hence one goes through a training program to adapt to the needs of the particular company or industry.

What are other main types of paths/options for someone who wants to make a career in embedded design ?

Outside of product-based embedded design, there is of course the semiconductor industry which manufactures chips. They design chips for all kind of products and then we can build on those basic chips to differentiate our product.

The other career path one can take is in the even more specialised field of electronic design automation tools — the tools that are used to manufacture chips.

What drove your choice of this career? Please give a glimpse of your journey to the current position?

It was mostly a sum of coincidence and interest. I entered the technology industry as a software engineer with Infosys, and since I was a fresher I was open to options. The firm had a small production engineering development division. I joined it and began liking the product engineering work. But since Infosys is largely a services company, the product development did not have much growth and hence to grow in the product development vertical I moved to Samsung.

What are the kinds of skills required to enter this industry?

It helps if you have studied either electronic and communications engineering or computer sciences. It is good to have strong basics in computer sciences and then the training for the particular industry takes over. The training is taken care of by the firms that hire you.

In your view, how crucial are the following in influencing one's path/growth in this career – does it matter which college you are from?

It really doesn't matter. Once you are hired, it basically depends on your output after the industry training.
– How much does my performance in college matter? Again, to get in the best firm, one should be a good scorer. But beyond that it depends on your output in the job.
– Are there some other certifications that can help? Maybe an advanced VLSI (Very Large Scale Integration) certification can help. The Indian Institute of Sciences offers a number of short term courses in different aspects of embedded design.

What are the unique highs and lows in this career?

One has constantly to keep improving knowledge, and that's the real challenge. My projects keep changing and the maximum we are on a project is for 6-8